Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 10 Handout

Thursday, October 15

Problem 10.1

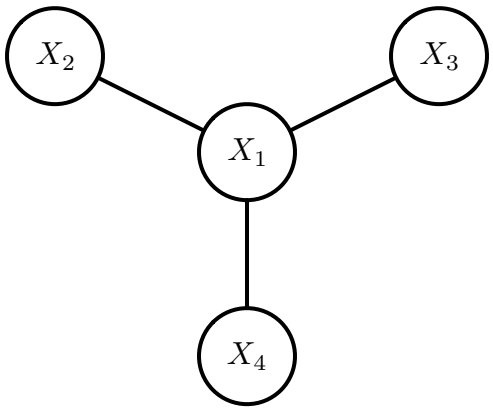
I have two coins, one fair (`F`) and one biased (`B`). Suppose I keep picking a coin and tossing it. You get to observe whether the coin I tossed landed heads (`H`) or tails (`T`). Your goal is to guess the sequence of coins I used from the observed coin toss results.

For the first toss, I choose either the fair coin or the biased coin uniformly at random. I prefer not to switch coins too often: after each toss, I use the same coin with probability 3/4 and switch coins with probability 1/4, independently of all other tosses. The fair coin is equally likely to land heads or tails. The biased coin lands heads with probability 1/4 and tails with probability 3/4.

Determine the most likely sequence of coin choices as you observe `THT`.

Problem 10.2

Let $X_1, X_2, X_3, X_4 \in \mathcal{X} = \{0, 1\}$ be binary random variables and consider the following tree graphical model representing the structure in the joint distribution p_{X_1,X_2,X_3,X_4}:



Moreover, suppose

$$\phi_{1,2}(x, x') = \phi_{1,3}(x, x') = \phi_{1,4}(x, x') = \begin{cases} 3 & x = 0,\ x' = 1 \\ 1 & \text{otherwise,} \end{cases}$$

$$\phi_2(x) = \phi_3(x) = \phi_4(x) = 1, \quad x \in \mathcal{X}, \quad \text{and} \quad \phi_1(x) = \begin{cases} \alpha & x = 1 \\ 1 & x = 0, \end{cases}$$

where $\alpha > 0$ is a constant. Compute all max-product algorithm messages and express the resulting most probable configuration (MPC) as a function of α.

Problem 10.3

Suppose you send your robot, Inquisitive, to Mars. Unfortunately, it gets stuck in a canyon while landing and most of its sensors break. You know the canyon has 3 areas. Areas 1 and 3 are sunny and hot, while Area 2 is cold. The temperature sensor still works. Every hour, you get a binary reading telling you whether the robot's current environment is `hot` or `cold`. The robot is equally likely to have landed in each of the three areas. Every hour, the robot tries to move forward by one area (i.e., from Area 1 to Area 2, or from Area 2 to Area 3). It succeeds with probability 3/4 and fails with probability 1/4. If it fails, it stays where it is. If it is in Area 3, it stays and waits to be rescued.

(a) Construct a hidden Markov model for this problem. Specify the initial state distribution, transition distribution, and the observation distribution.

(b) Suppose we observe the sequence (`hot`, `cold`, `hot`). Before doing any computation, determine the sequence of locations that produced these observations. Then compute the distribution of the second state using message passing. Do your answers match up?